Luck, Competitve Balance, Planning

October 25, 2020

Overall 100pt. Submit both markdown and pdf files

This homework consists of two parts. Where part one is traditional one, while part two is aimed to support your individual and group projects. In order to guarantee decent projects, we have one week to build roadmap of your projects and start the implementation stage.

Please try to build your roadmap as comprehensive as possible and remember: **problem well defined is problem solved.**

Part 1

In this homework you should use the functions you created during previous homework: **Final Standings**.

Skill, Luck? (25p)

Your are assigned to estimate Luck and Skill for previously assigned soccer ligues.

1. Calculate contribution of luck per season. (15p)
2. Visualize the trend over season. (5p)
3. Interpret the visualization. (5p)

Concentration Index (25p)

1. Calculate concentration index per season for your ligues:from CI2 to CI6. Assuming game outcome can only be **W** or **L**. (15p)
2. Visualize CIs and interpret your findings. (5p)
3. Is there significant difference between CIs? (5p)

Part 2

Individual Project (20p)

Here you should elaborate detailed steps of your **Individual Project**. If you have questions regarding the steps please include them as well.

Group Project (30p)

Here you should elaborate detailed steps of your **Group Project**. If you have questions regarding the steps please include them as well.